

Doctor Hub - Final Semester Project Documentation

Project Overview

Doctor Hub is a healthcare consultation and patient history management system where patients can search doctors according to disease and treatment type including Allopathic, Homeopathic, and Herbal doctors.

Main Features

- Doctor search and filtering
- Appointment booking system
- Medical history sharing
- Prescription management
- Payment verification system
- Doctor assistant management
- Clinic and schedule management
- Patient-doctor communication system

User Roles

Patient: Book appointments, manage history, upload reports.

Doctor: Add prescriptions, manage schedules and clinics.

Assistant: Verify payments and bookings.

Admin: Manage doctors and users.

Super Admin: Full system control.

Authentication Features

- User Registration
- Login System
- Forgot Password
- JWT Authentication
- Role-Based Access Control

Medical History Rules

- Medical history cannot be deleted.
- Doctors can only add new records.
- Previous prescriptions cannot be edited.
- Patients cannot remove doctor prescriptions.

Appointment Workflow

1. Patient searches doctor.
2. Patient filters doctor according to disease.
3. Patient books appointment.
4. Payment screenshot uploaded.
5. Assistant verifies payment.
6. Appointment confirmed.

Database Tables

- users
- doctors
- patients
- appointments
- prescriptions
- medical_history
- payments
- assistants
- clinics

REST APIs

POST /api/auth/register
POST /api/auth/login
GET /api/doctors
POST /api/appointments
POST /api/payments
GET /api/history

Security Features

- JWT Authentication
- Encrypted Passwords
- Validation Middleware
- Secure APIs
- Protected Medical Records

Future Enhancements

- AI disease prediction
- Video consultation
- WhatsApp notifications
- E-prescription PDF generation

Marks Distribution

Module	Marks
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Architecture Design	15
Database Design	15
Authentication & RBAC	10
Workflow Logic	15
API & Backend	10
Frontend UX	10
Analytics & Reports	10
Code Quality	5
Deployment	5
Viva & Presentation	5

Conclusion

Doctor Hub is a production-style healthcare consultation platform demonstrating advanced workflow management, appointment handling, payment verification, RBAC implementation, and secure medical history management.